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EXPERIMENT 7

Aim: Creating VM on Google Cloud Platform.

Theory:

Step 1: Sign in to Google Cloud Console

1. Go to [Google Cloud Console](#).
2. Sign in with your Google account.
3. If this is your first time, set up a new project or select an existing project.

Step 2: Navigate to the VM Instances Page

1. In the left sidebar, click on "**Compute Engine**" under the "Compute" section.
2. Then, click on "**VM instances**".

Step 3: Create a New VM Instance

1. On the **VM Instances** page, click the "**Create**" button at the top of the page.

Step 4: Configure the VM Instance

Fill in the configuration options for your VM instance:

1. Name:

- Enter a name for your VM instance (e.g., my-vm-instance).

2. Region and Zone:

- Select the region and zone where you want your VM to be hosted. The options are available in the dropdown.

3. Machine Type:

- Select the machine type for the VM. Choose an appropriate configuration based on your needs (e.g., e2-medium or n1-standard-1).

4. Boot Disk:

- Click on "**Change**" to select the OS for your VM.
- You can choose a public image (e.g., Ubuntu, CentOS, or Windows), or if you need a custom image, you can upload one.



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- Set the disk size according to your requirements (e.g., 10 GB or higher).

5. Firewall:

- Check "**Allow HTTP traffic**" and "**Allow HTTPS traffic**" if you want your VM to serve websites or web applications.

Step 5: (Optional) Set Up Additional Options

1. **Identity and API Access:** You can specify the service account and API access scopes based on your application needs.
2. **Networking:** Configure network settings (e.g., setting a static IP, enabling private IPs, etc.).
3. **Disks and GPUs:** If you need additional disks or GPUs, you can add them in this section.
4. **Startup Script:** If you want to automate any configuration tasks on VM startup, you can provide a startup script here.

Step 6: Create the VM

Once you've configured everything, click on the "**Create**" button at the bottom to launch your VM.

Step 7: Access the VM

1. After the VM is created, it will appear in the list of VM instances.
2. To connect to your VM:
 - Click on the **SSH** button next to your VM instance. This will open an SSH connection in your browser.

Screenshots:

Step 1: Create instance



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Connect

VM Instances

Compute Engine lets you use virtual machines that run on Google's infrastructure. Create micro-VMs or larger instances running Debian, Windows, or other standard images. Create your first VM instance, import it using a migration service, or try the quickstart to build a sample app.

CREATE INSTANCE

TAKE THE QUICKSTART

Step 2: Configure machine



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Create an instance

CREATE VM FROM...

Machine configuration

c2-standard-4, us-west1-a

OS and storage

Debian GNU/Linux 12 (bookworm)

Networking

1 network interface

Observability

Install Ops Agent

Security

Advanced

Machine configuration

Name *

cclpractical

Region *

us-west1 (Oregon)

Zone *

us-west1-a

Region is permanent

Zone is permanent

General purpose

Compute optimized

Memory optimized

Storage optimized

GPUs

Machine types for performance-intensive workloads, with highest performance per core

Series	Description	vCPUs	Memory	CPU Platfr
☐ H3	High performance computing workloads	88	352 GB	Intel Sapph
☒ C2	Ultra-high performance, compute-intensive workloads	4 - 60	16 - 240 GB	Intel Casc
☐ C2D	Ultra-high performance, compute-intensive workloads	2 - 112	4 - 896 GB	AMD Millar

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads.

c2-standard-4 (4 vCPU, 2 core, 16 GB memory)

vCPU

4 (2 cores)

Memory

16 GB

ADVANCED CONFIGURATIONS

Monthly estimate

\$122.94

That's about \$0.17 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
4 vCPU + 16 GB memory	\$152.43
10 GB balanced persistent disk	\$1.00
Use discount	-\$30.49
Logging	Cost varies
Monitoring	Cost varies
Total	\$122.94

Compute Engine pricing

Cloud Operations pricing

LESS

Machine configuration

Some form fields are incorrect

OS and storage

Debian GNU/Linux 12 (bookworm)

Networking

1 network interface

Observability

Install Ops Agent

Security

Advanced

Machine configuration

Name *

cclpractical

Region *

us-west1 (Oregon)

Zone *

us-west1-a

Region is permanent

Zone is permanent

General purpose

Compute optimized

Memory optimized

Storage optimized

GPUs

Machine types for workloads with higher memory-to-vCPU ratios, like in-memory databases

Series	Description	vCPUs	Memory	CPU Platform
☐ X4	Extra Large, in-memory databases	960 - 1920	16,384 - 32,768 GB	Intel Sapphi
☐ M3	High memory, memory-intensive workloads	32 - 128	976 - 3,904 GB	Intel Ice Lak
☒ M2	Ultra-high memory, in-memory databases	208 - 416	5,888 - 11,776 GB	Intel Cascad
☐ M1	High memory, memory-intensive workloads	40 - 160	961 - 3,844 GB	Intel Skylake

M2 isn't available in us-west1-a

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads.

m2-megamem-416 (416 vCPU, 208 core, 5,888 GB memory)

vCPU

416 (208 cores)

Memory

5,888 GB

ADVANCED CONFIGURATIONS

You might not have enough quota for this VM

Try a different configuration or request an increase with a quota adjustment.

Quota for	Current usage/limit	Limit needed
us-west1	M2 CPUs	0/0 416

REQUEST QUOTA ADJUSTMENT

LESS

Monthly estimate

\$36,713.67

That's about \$50.29 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
416 vCPU + 5,888 GB memory	\$36,712.67
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Total	\$36,713.67

Compute Engine pricing

Cloud Operations pricing

LESS

CREATE

CANCEL

EQUIVALENT CODE



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Create an instance

CREATE VM FROM...

Machine configuration

e2-medium, us-west1-a

OS and storage

Debian GNU/Linux 12 (bookworm)

Networking

1 network interface

Observability

Install Ops Agent

Security

Advanced

us-west1 (Region)

us-west1-a

Region is permanent

Zone is permanent

General purpose

Compute optimized

Memory optimized

Storage optimized

GPUs

Machine types for common workloads, optimized for cost and flexibility

Series	Description	vCPUs	Memory	CPU Platform
C4	Consistently high performance	2 - 192	4 - 1,488 GB	Intel Emerald Rapids
C4A	Arm-based consistently high performance	1 - 72	2 - 576 GB	Google Axion
N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald Rapids
C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel Sapphire Rapids
C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Intel Broadwell

Filter instance sizes

Shared-core

Standard

High memory

High CPU

e2-micro

e2-small

e2-medium

vCPU

1-2 vCPU (1 shared core)

Memory

4 GB

ADVANCED CONFIGURATIONS

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Total	\$25.46

Compute Engine pricing

Cloud Operations pricing

LESS

Step 3: Choose machine type

Create an instance

CREATE VM FROM...

Machine configuration

e2-micro, us-west1-a

OS and storage

Debian GNU/Linux 12 (bookworm)

Networking

1 network interface

Observability

Install Ops Agent

Security

Advanced

us-west1 (Region)

us-west1-a

Region is permanent

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C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Intel Broadwell
N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade Lake
N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD Milan
T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra
T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMD Milan
N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB	Intel Haswell

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

PRESET

CUSTOM

e2-micro (2 vCPU, 1 core, 1 GB memory)

vCPU

0.25-2 vCPU (1 shared core)

Memory

1 GB

ADVANCED CONFIGURATIONS

Monthly estimate

\$7.11

That's about \$0.01 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 1 GB memory	\$6.11
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Total	\$7.11

Compute Engine pricing

Cloud Operations pricing

LESS



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External IP

Connect

☐

cdpractical

us-west1-a

SSH

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Explore Backup and DR

Back up your VMs and set up disaster recovery

View billing report

View and manage your Compute Engine billing

Monitor VMs

View outlier VMs across metrics like CPU and network

Explore VM logs

View, search, analyze, and download VM instance logs

Set up firewall rules

Control traffic to and from a VM instance

Patch management

Schedule patch updates and view patch compliance on VM instances

Load balance between VMs

Set up Load Balancing for your applications as your traffic and users grow

Conclusion:

Thus we successfully Created VM on Google Cloud Platform.